

Lab 10: Modelling epidemics and endemic breakouts of infectious diseases

Krutarth Kadia (202201475)* and Het Shah (202201515)†
*Dhirubhai Ambani Institute of Information & Communication Technology,
Gandhinagar, Gujarat 382007, India
CS302, Modeling and Simulation*

This study examines the dynamics of epidemics and endemic diseases within populations, utilizing mathematical models to analyze the spread and persistence of infectious diseases. Through numerical simulations and graphical representations, we explore the behavior of key variables such as infected, susceptible, and recovered individuals over time, shedding light on the factors influencing disease outbreaks and endemic stability.

I. EPIDEMICS

B. Solution

A. Model Equations

The coupled dynamics of the variables are given by:

$$\begin{aligned}\dot{x} &= Axy - Bx \\ \dot{y} &= -Axy \\ \dot{z} &= Bx\end{aligned}$$

where A is the infection rate and B is the removal rate ($A, B > 0$).

Value of $A = 2.18 \times 10^{-3}$ value of $B = 0.44$

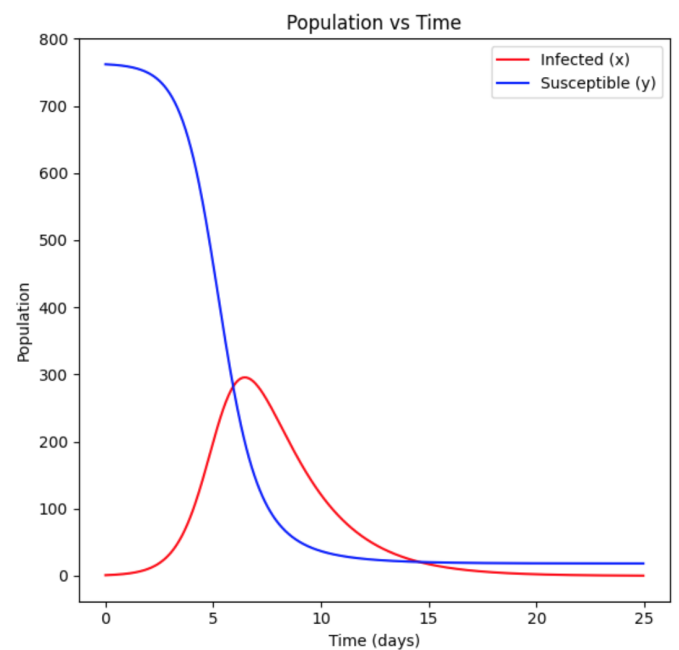


FIG. 1: Plot of x and y with respect to time t (without logarithmic scale)

*Electronic address: 202201475@daiict.ac.in

†Electronic address: 202201515@daiict.ac.in

Max value of x reach at time $t = 6.5$ days

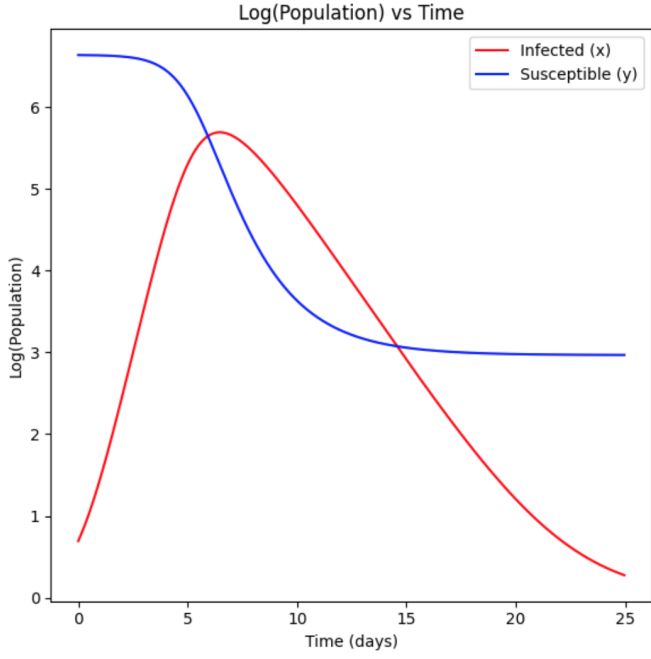


FIG. 2: Plot of x and y with respect to time t (with logarithmic scale)

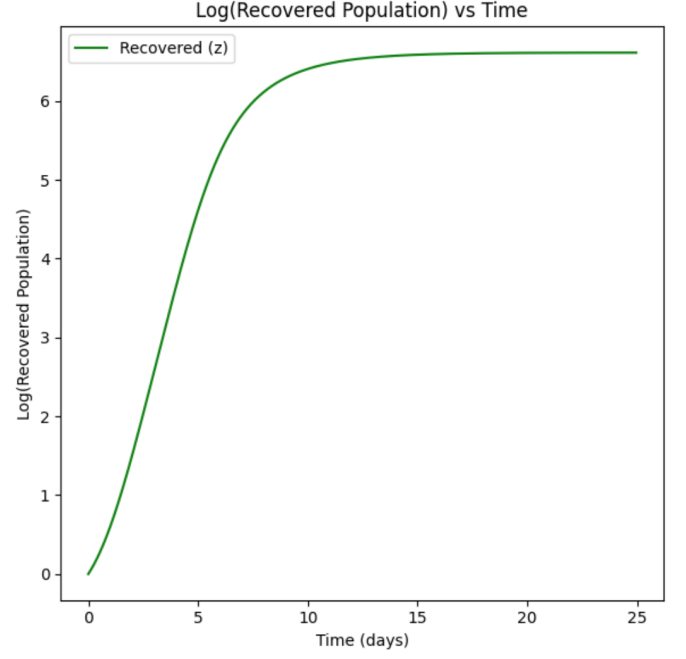


FIG. 4: Plot of z with respect to time t (with logarithmic scale)

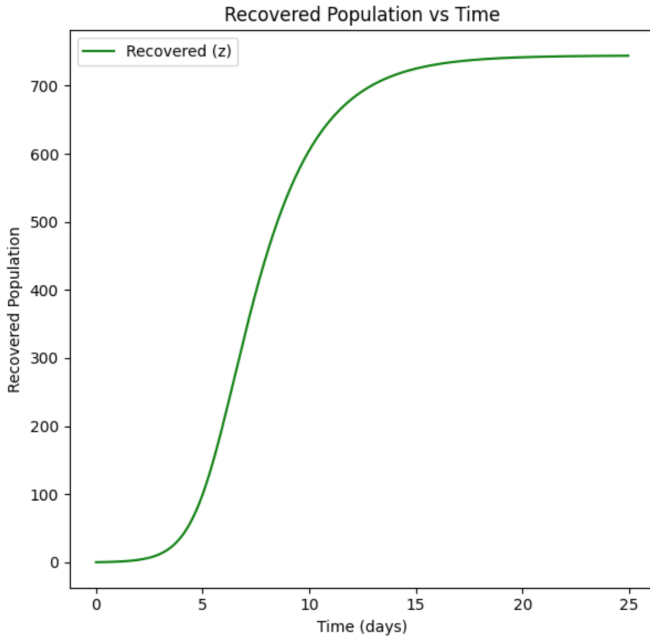


FIG. 3: Plot of z with respect to time t (without logarithmic scale)

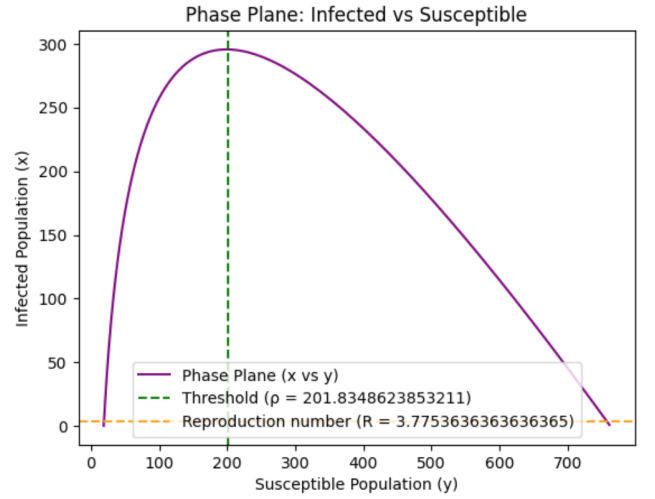


FIG. 5: Plot of x vs y

II. ENDEMIC DISEASES

A. Model Equations

The coupled system of equations is given by:

$$\dot{x} = Axy - Bx - ax$$

$$\dot{y} = bN - Axy - ay$$

$$\dot{z} = Bx - az$$

$$\dot{N} = (b - a)N$$

where $N \equiv N(t)$, a is the per capita death rate, and b is the per capita birth rate ($a, b > 0$).

$$a = b = 0.02 \text{ year}^{-1}$$

$$A = 10^{-6} \text{ year}^{-1}$$

$$B = 0.333 \text{ year}^{-1}$$

$$N = 10^6$$

$$x_0 = 10^5$$

$$y_0 = 9 * 10^5$$

B. Solution

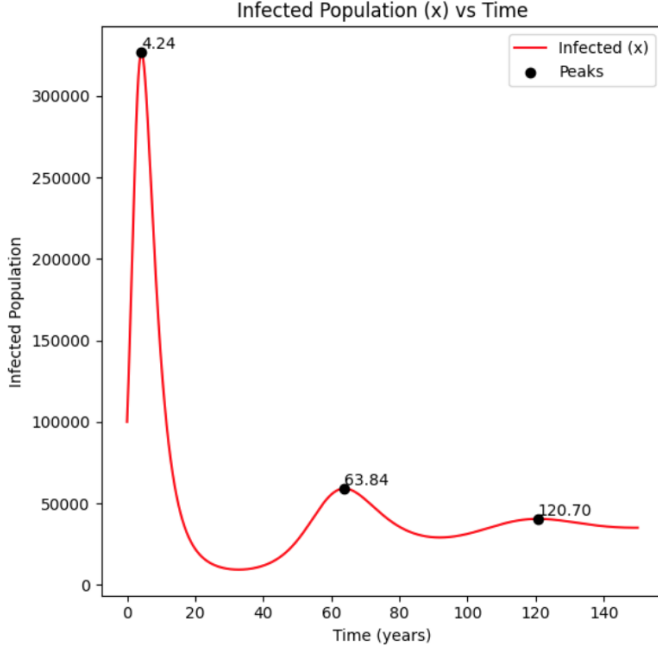


FIG. 6: Plot of x with respect to time t

The Endemic occurs at 4.2438 year , 63.8438 year , 120.7014 year in a span of 150 years.

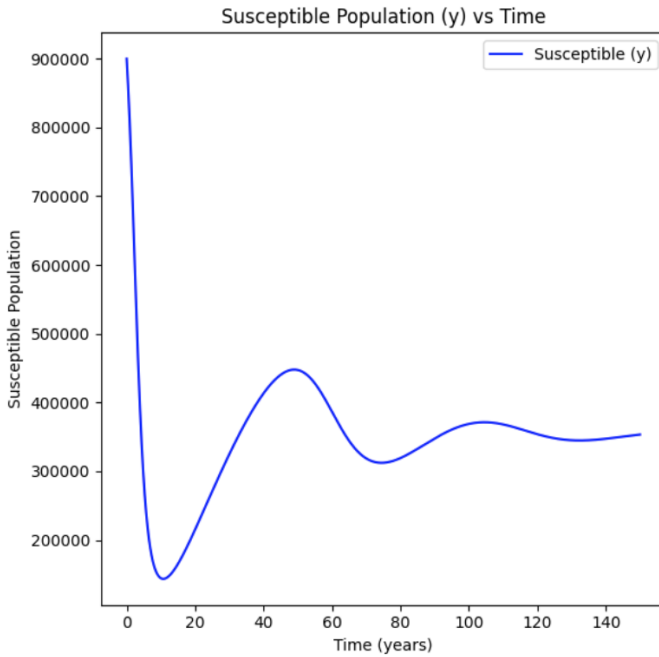


FIG. 7: Plot of y with respect to time t

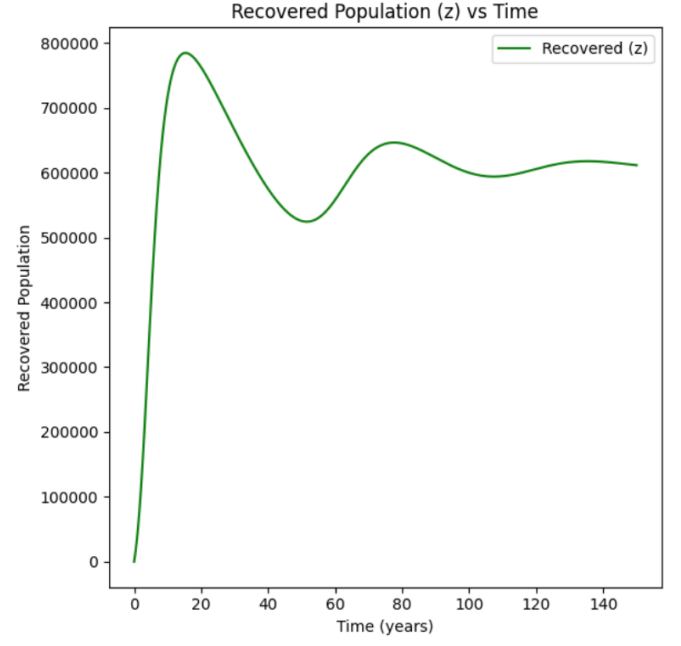


FIG. 8: Plot of z with respect to time t